Secure Financial Corporation Bank (SFC)

Project Brief

INTRODUCTION

SFC Bank (Secure Financial Corporation) is a mid-sized banking institution in Bahrain offering retail banking, internal corporate services, and online banking solutions. To support these operations, the bank requires a secure, reliable, and scalable network infrastructure capable of maintaining efficient communication between its Headquarters (HQ) and a remote branch office.

The proposed network architecture follows a hierarchical design, with the Headquarters hosting critical services. These include a centralized server providing Active Directory, RADIUS-based AAA authentication, and Syslog services for monitoring and auditing network activity. A dedicated Demilitarized Zone (DMZ) is also implemented to host the bank's public-facing web server, ensuring isolation from the internal network and enhancing overall security.

Both the Headquarters and branch office are protected by firewalls that enforce strict traffic control between internal and external networks. Inter-site connectivity is achieved through a dual WAN design, utilizing an MPLS Layer 3 VPN as the primary link and a Site-to-Site VPN over the public ISP as a backup. This approach ensures secure, reliable, and encrypted communication, with automatic failover in case of link failure.

The objective of this project is to design and implement a secure, high-availability enterprise network that delivers strong perimeter defense, resilient inter-branch connectivity, centralized authentication, and effective network segmentation.

The solution will be developed and tested in a virtualized environment using EVE-NG, enabling realistic simulation of enterprise network deployment and validation of security and failover mechanisms.

PROJECT DESCRIPTION

This project focuses on designing and implementing a secure, enterprise-grade network infrastructure for SFC Bank. The proposed solution connects the Headquarters (HQ) with a remote branch office while ensuring strong security, high availability, and proper network segmentation.

The network follows a hierarchical architecture, where the Headquarters hosts critical internal services. These include a centralized server providing Active Directory, RADIUS-based AAA authentication, and Syslog services for monitoring and auditing network activities.

A dedicated Demilitarized Zone (DMZ) is deployed at the Headquarters to host the bank's public-facing web server. This design isolates external services from the internal network, reducing the risk of unauthorized access and improving overall security.

Both the Headquarters and branch office are secured using firewalls that control and filter traffic between internal and external networks. Inter-site communication is established using a dual WAN approach: an MPLS Layer 3 VPN as the primary connection and a Site-to-Site VPN over the public ISP as a backup. This ensures encrypted communication, reliable connectivity, and automatic failover in case of link failure.

The entire network is implemented in a virtualized lab environment using EVE-NG, allowing realistic simulation of enterprise network deployment and testing of security and failover mechanisms.

Objectives

The main objectives of this project are to:

• Design a secure enterprise network architecture suitable for a banking environment.

• Provide reliable connectivity between the Headquarters and branch office using secure communication methods.

• Implement network segmentation through VLANs and a dedicated DMZ to isolate public services from internal systems.

• Deploy centralized authentication and authorization using Active Directory and RADIUS-based AAA services.

• Implement centralized logging using Syslog to monitor network activity and administrative access.

• Protect internal networks using firewall security policies.

• Secure inter-branch communication using a Site-to-Site VPN tunnel over the ISP infrastructure.

• Implement MPLS Layer 3 VPN for primary inter-branch communication.

• Provide automatic failover between MPLS and Site-to-Site VPN to ensure high availability.

• Ensure uninterrupted banking services during WAN link failures.

• Simulate a real-world enterprise network environment using EVE-NG.

Benefits

The implementation of this network architecture provides several benefits for SFC Bank:

• Improved Security: Firewalls, VPN encryption, and network segmentation help protect sensitive banking data from unauthorized access.

• Secure Inter-Branch Communication: The Site-to-Site VPN ensures that communication between HQ and the branch is encrypted and protected from interception.

• Centralized Authentication: Active Directory and AAA services allow centralized control of user access and administrative privileges.

• Network Monitoring and Auditing: Syslog logging enables administrators to track network activity and identify potential security issues.

• Isolation of Public Services: Hosting the web server in the DMZ prevents direct access to internal banking systems from the internet.

• Scalability: The hierarchical design allows the network to expand easily by adding additional branches or services in the future.

• High Availability: Automatic failover ensures continuous connectivity between HQ and branch.

• Performance Optimization: MPLS provides stable and low-latency communication for critical banking applications.

• Redundancy: Backup VPN ensures service continuity during MPLS outages.

• Realistic Simulation: Implementing the network in EVE-NG provides a realistic testing environment that simulates real enterprise networking deployments.

SYSTEM REQUIREMENTS

- 50 GB RAM
- 30 vCPUs
- 512 GB storage SSD
- Cisco IOSv router and L2/L3 switch images
- Cisco ASAv Image.
- Windows Server 2019, Windows 10 images.

PROJECT PLAN

Task	Estimated Duration	Start Date	End Date
Information and Requirements Gathering	2 weeks	14/2/2026	22/2/2026
Planning	3 weeks	23/2/2026	8/3/2026
Execution	6 weeks	9/3/2026	19/4/2026
Testing	1 week	20/4/2026	27/4/2026
Closing	5 days	28/4/2026	5/5/2026

MS project will be used to monitor the progress of the project and manage every major part.

PROPOSED TECHNOLOGIES

The chosen technologies for this project were selected because they meet industry requirements for secure wide area networking, are suitable for a distributed multinational setting, and are applicable to current enterprise network design.

EVE-NG: designing and testing network configurations and scenarios. It was selected Because it is the only platform that can handle the entire SFC system without collapsing.

Routers & Switches: for the network infrastructure.

Firewall: to control and filter incoming and outgoing traffic between internal networks and external connections.

Windows Server: hosting various network services and resources.

PuTTY: A terminal emulator for secure remote access to network devices, used for configuration and management tasks.

Endpoint devices: such as computers that connect to the network, including Windows 10 PCs.